Supplementary Material

This is the supplement to "Only Means Only: what a harmonization benchmark score averages over". Main-text numbers are cited as "main Section N" or "main Table N", and the figures and tables here are numbered S1 onwards. Values are measured from saved run records unless the text calls them registered design values, the budgets and predictions fixed before a measurement.

S1 Graded exposure

The dose k is the number of train composites built from the same target photograph as a test composite. The dose distribution over the 7,404 official test composites is $k = 0$: 5,582, $k = 1$: 1,024, $k = 2$: 443, $k = 3$: 161, $k = 4$: 103, $k = 5$: 45, $k = 6$: 12, $k = 7$: 25, $k = 8$: 8, $k = 9$: 1. The $k = 0$ count is 5,582 rather than 5,581 because one of the 1,823 excluded composites was excluded as a decoded-pixel duplicate (its target is pixel-identical to a train target under a *different* filename), so its identity string never occurs in the train list and the dose counter correctly assigns it $k = 0$. The remaining 1,822 exclusions are path-identity matches and carry $k \geq 1$, which is exactly the sum of the $k \geq 1$ counts above.

Exposure step (MSE ↓)	Harmonizer	DoveNet
Pooled, unmatched	−106.98	−70.90
Per-bin steps, sign	8/8 negative	8/8 positive
Per-bin step range	−293.7 to −0.3	+0.9 to +69.2

Table S1: Graded exposure dose-response for both baselines. Steps are $k \geq 1$ minus $k = 0$ inside foreground-ratio bins, so a negative entry means the shared composite scores better. The bin-averaged step and its detection floor are in main Table 2.

Method (MSE ↓)	ret.	excl.	offic.	Δ
	2,476	1,807	n/a	excl.−ret.
HCOCO only, the population of the HCOCO-only matched analysis (main Table 2)				
DoveNet	98.96	67.16	n/a	−31.80
Harmonizer	40.24	24.92	n/a	−15.32
Editor (ours)	74.11	47.00	n/a	−27.12

Table S2: HCOCO-only score rows for the same three methods; the pooled version is main Table 1 and the matched, foreground-controlled version is main Table 2.

S2 Fixed-model probe

At five seeds the smallest attainable two-sided sign-test value is $p = 0.0625$, and the in-region L1 gap is positive in all five seeds (+1.954 to +2.938).

S2.1 Resampling unit of the probe intervals

The bottom panel of Table S3 reports intervals from a cluster bootstrap whose resampling unit is the target photograph, so every composite built on one photograph moves with it. The committed probe script originally resampled composites within each foreground-ratio bin, which ignores that correlation and returns narrower intervals. The script now implements the photograph unit

	Excluded	Retained	Contrast
Unmatched, full populations (1,823 / 5,581)			
In-region L1 ↓	22.58	25.01	5/5 seeds
MSE ↓	48.82	149.45	3.06×
fMSE ↓	871.77	1104.82	1.27×
PSNR ↑ (dB)	35.04	31.91	−3.13
Do-nothing L1 ↓	24.05	29.64	1.23×
Within HCOCO: unmatched → foreground-matched [95% CI]			
MSE	−27.12		−2.25 [−10.09, +6.20]
fMSE	−52.45		−42.77 [−97.12, +15.56]
PSNR (dB)	+1.30		+0.04 [−0.22, +0.30]

Table S3: The fixed-model probe. Top: five per-seed checkpoints scored on the two halves of the official list, every contrast retained relative to excluded, the in-region gap positive in all five seeds. Bottom: excluded-minus-retained differences inside HCOCO.

as its default, and on the recorded seed and configuration it reproduces the reported intervals to within bootstrap noise:

Interval	Reported	Re-run, photograph clusters	Re-run, composite clusters
MSE	[−10.09, +6.20]	[−9.92, +5.50]	[−5.43, +1.25]
fMSE	[−97.12, +15.56]	[−96.43, +12.52]	[−66.25, −19.01]
PSNR (dB)	[−0.22, +0.30]	[−0.22, +0.31]	[−0.06, +0.15]

Table S4: The fixed-model probe intervals under the two resampling units, 2,000 draws each, 8 foreground-ratio bins, seed as recorded. The photograph unit reproduces the reported figures; the composite unit does not, and it is the unit that matters here, because a test composite is correlated with every training composite built on the same photograph.

The residual differences between the reported and re-run photograph-clustered intervals are bootstrap noise: 2,000 draws over 3,417 photograph clusters. The choice of unit is not cosmetic. Under the composite unit the matched fMSE interval excludes zero, so a reader who took that unit would read a detected difference where the photograph unit reads none. That is the reason the unit is stated here and in the main text: the paper's argument is that composites sharing a photograph are not independent, and the uncertainty calculation has to use the unit the audit keys on.

S3 Three-population scores for every checkpoint

In every pooled bin 92.5 to 100 percent of the exposed composites are HCOCO, while the HCOCO share of the unexposed rises from 0.52 in the first bin to 0.76 in the second and then falls to 0.22 in the eighth (largest-foreground) bin, and the two largest-foreground bins carry 79, 82 and 80 percent of the bin-averaged pooled step for the three methods.

S4 Near-duplicate audit

A 64-bit dHash is computed over every unique real background image: HCOCO 26,912 train against 2,025 retained-test, HAdobe5k 3,897 against 432, HFlickr 4,397 against 436, Hday2night

	DoveNet	Harmonizer	PCT-Net	DucoNet	DiffHarmony
MSE, official	122.75	107.08	78.56	107.60	—
fMSE, official	1132.68	794.24	608.26	791.32	—
MSE, retained	140.18	133.42	97.41	135.04	162.42
fMSE, retained	1114.71	894.34	682.00	905.86	523.90
MSE, excluded	69.40	26.43	20.83	23.58	196.67
fMSE, excluded	1187.67	487.80	382.50	440.63	512.95
retained/excluded	2.02	5.05	4.68	5.73	0.83

Table S5: Every checkpoint on the same three lists, one protocol, so the official column is the pooled number and the other two are the populations composing it. DiffHarmony has no official row, its bundle not having been re-extracted, and is reported descriptively.

Method	Population	MSE	fMSE	composites
Editor (five seeds)	retained	149.45	1104.82	27,905
Editor (five seeds)	excluded	48.82	871.77	9,115

Table S6: The in-house typed editor, five per-seed checkpoints scored once each, for reference. It is a control in this study and not a harmonization-quality claim; its bound carries slack and is below the edit magnitude for two of the five seeds (Section S15).

82 against 24. Test is queried against train at Hamming radii $\{0, 2, 4, 6, 8\}$, and every Hamming-0 pair is confirmed by a 64×64 grayscale correlation.

Hamming radius	Pairs	Status
0	4	confirmed (1 HCOCO, 1 HAdobe5k, 2 HFlickr; three with $r \geq 0.9999$, one with $r = 0.9509$)
≤ 2	5	candidate pairs
≤ 4	8	candidate pairs
≤ 6	19	candidate pairs
≤ 8	78	candidate pairs

Table S7: Cross-split pairs beyond the registered exclusions. Wider radii retrieve unadjudicated candidates that mix same-scene and re-encoded variants: neither the candidate counts nor the four confirmed pairs bound total residual sharing from above. The confirmed floor is 0.14 percent.

Whole-matrix screen. The hash radii above are bounded by the radii tested. A second, independent pass compares every one of the 7,404 test composites with every one of the 65,742 training and validation composites at 32×32 grayscale correlation and reports cross-split pairs above 0.99 whose identity strings differ. It returns six candidates. One is byte-identical and its composite is one of those removed by the decoded-pixel check. The other five sit in the retained list at correlation 0.99997 or above; four are composites of three of the four confirmed pairs above, and only the fifth, a screen candidate rather than a confirmed pair, is additional. The screen is a correlation at a fixed scale, so it is a floor on what a stricter comparison would find, not a proof of cleanliness.

S5 External comparison on ccHarmony

Provenance of the numbers above. The probe's excluded population reads 4.63 against the retained one, and the within-HCOCO interval is the one in the committed record: the committed probe script reproduces its point estimates and not this interval.

Descriptive quantity	Value
Retained sign agreement	5 of 6
Official sign agreement	6 of 6
$\Delta_{\text{agreement}}$	-1 of 6
LOMO without DucoNet: Delta sign	0
LOMO without Harmonizer: Delta sign	0
Kendall tau (retained, ccHarmony)	0.6667
95% interval	$[0.6667, 1.0]$

Table S8: External comparison (E2, the ccHarmony comparison of main Section 4.5): descriptive quantities within the frozen four-method roster. Signs use point gaps. Two leave-one-method-out removals change the full-roster Delta sign to zero. This table carries no efficacy verdict or external-validation claim.

S6 Gate evidence and training configuration

The second factor's capacity search overshoots the first by 2,592 active parameters (47,235 against 44,643), a width confound the paired ablation is free of by construction.

The editor is a compact CNN on 256×256 inputs, run at $(\gamma_P, \gamma_S) = (0.1, 0)$. Five seeds $\{0, 1, 2, 3, 4\}$, halo 0, 15 epochs, $N = 1024$, batch 16, best-on-validation checkpointing, strict cuDNN determinism in paper mode: the control arm measures $|\Delta| = 1.849$ against the treatment at exactly 0.000.

Quantity (held-out validation L1)	Value
Typed channel vs capacity floor	$+0.358$ $[+0.071, +0.645]$
seeds positive	5/5
Paired γ_P ablation (5 seeds)	$+0.694$ $[+0.510, +0.878]$
seed-to-seed sd	0.148
two-sided sign test	$p = 0.0625$
Oracle signal, F1 $-$ F2	$+0.994$ $[+0.856, +1.132]$
Price of certification, F3 $-$ F2	$+0.636$ $[+0.451, +0.822]$
Certified share of oracle signal	≈ 36 %
Physical certificates (range)	0.049 to 0.487
Semantic certificates	exactly 0

Table S9: Gate evidence and the three-factor decomposition, five seeds on held-out validation. The paired ablation is the pre-registered primary estimator: identical shapes, initialization and batches, with only the physical gate differing, so no selection enters it.

Tightness: the held-out projected-gradient check (100 steps, $\varepsilon = 8/255$, 100 inputs for each of the five seeds, 500 searches, zero soundness violations) gives per-seed ratios of mean achieved change to mean bound of 0.070, 0.077, 0.094, 0.073, 0.072 (mean 0.077, worst single input 0.28). Table S22 gives the physical-certificate bound at $\varepsilon = 8/255$ for all five selected checkpoints. It ranges from 1.85 to 17.70 levels.

S7 The third method in the standard protocol

Provenance and record names. The two public baselines are re-derived from pinned public code and checkpoints in the re-derivation run `20260830-174720-baseline-rederivation`, over the retained list `retained_5581`.

Subset	n	MSE ↓	fMSE ↓	PSNR (dB) ↑
Editor (ours, certified, composited), 5 seeds, once-scored				
overall	5,581	149.45 [146.92, 151.98]	1104.82 [1069.52, 1140.12]	31.91 [31.74, 32.07]
HCOCO	2,476	74.11	919.69	33.79
HAdobe5k	2,160	196.42	1178.66	30.81
HFlickr	812	263.56	1461.59	28.59
Hday2night	133	92.60	1173.86	34.98
in-region L1 25.01 [24.45, 25.56]; no-op source in-region L1 29.64				
DoveNet (re-derived)				
overall	5,581	140.18	1114.71	32.37
HCOCO	2,476	98.96	1240.67	32.69
HAdobe5k	2,160	112.82	701.43	33.61
HFlickr	812	344.74	1792.76	27.73
Hday2night	133	102.97	1342.01	34.63
Harmonizer (re-derived)				
overall	5,581	133.42	894.34	33.43
HCOCO	2,476	40.24	529.98	36.27
HAdobe5k	2,160	235.52	1294.19	31.01
HFlickr	812	159.61	967.61	30.69
Hday2night	133	50.33	736.13	36.64

Table S10: Standard-protocol metrics on the retained list, every method on the identical list. Our rows are the five-seed mean of the per-seed checkpoints and the two public baselines are re-derived from pinned code.

S8 Cross-dataset construction scan

Per-collection file counts. ccHarmony file level: 4,260 composites present (3,080 and 1,180), 250 train and 100 test real files referenced, none missing, no real file or mask shared across the split, 350 real files all referenced. DESOBA file level: zero missing files in four folders, 1,012 images equal to 839 + 160 + 13, and no CRC-identical image across the split. The DESOBA shared stem is `web-shadow0248`, whose test `-1` and train `-2` read as near-duplicate scene content at grayscale correlation 0.35.

The construction scans use a name-level, CPU-only, read-only identity matcher. The scan script is released in the project repository as `dataset_identity_scan.py`, and the counts below are recorded values from it. Identity rule: list entry without directory and extension, split on `_`. The last two tokens (variant letter or mask id, and composite index) are dropped. Identity is the first token, or the first two tokens joined when the name has four or more tokens. IHarmony4 identities are scoped by sub-dataset. DESOBA uses the file stem minus a trailing `-N` variant index.

Blocked or not scanned: iHarmony8K (no public source found after three bounded searches). DESOBAv2 (the only reachable archive is 12.69 GiB and was not fetched, and no split documentation was found). SRD, USR, WSID (no verified source). ISTD+ (not a composite benchmark, not scanned). HYouTube (not located, not scanned). The scan does not show whether identically named identities are the same photograph, near-duplicate or re-photographed content across splits, foreground overlap, leakage through real-image pools or pretraining data, or any effect on

Dataset (units)	Train / test units	Train / test identities	Shared	Test covered
ccHarmony, published lists (composites)	3,080 / 1,180	250 / 100	0	0 of 1,180
iHarmony4 official lists (composites)	65,742 / 7,404	35,288 / 4,317	1,399	1,822 of 7,404 (24.6%)
iHarmony4 retained lists (composites)	65,742 / 5,581	35,288 / 2,917	0	0 of 5,581
ccHarmony test vs iHarmony4 official train	1,180 test	100 / 35,288	0	0
DESOBA (source images)	839 / 160	816 / 160	1	1 of 160 (0.6%)

Table S11: Construction scan. The recorded iHarmony4 figure is 1,823 of 7,404, of which 1,822 are path-identity composites and one is a decoded-pixel duplicate. The DESOBA shared stem reads as a near-duplicate scene at grayscale correlation 0.35, and ccHarmony shares no real file with the test side.

model scores. The DESOBA pair counts (2,995 train and 624 test object-shadow pairs) are the README's and were not re-derived. The ccHarmony cross-check against iHarmony4 has low power because the name formats do not overlap.

S9 Provenance of the reported values

Audit. The four-checkpoint audit values (main Section 4.4) come from its retained evaluation record. The MSE and PSNR sensitivity reads come from the retained MSE, PSNR and metric-agreement records, kept with the sensitivity-grid record.

Retrain estimates and training check. The difference-in-differences and training-check values of main Table 3 come from a replacement record generated after a fresh, detached scoring pass of the training check (called M8 in the repository records). A canonical aggregate record named in the retrain plan does not exist. The replacement record is used in its place, and we do not claim the two would be interchangeable. An earlier M8 scoring directory is superseded and none of its values is used. The fresh directory holds twelve row files and twelve sidecar files. A first verification re-hashed the twelve scored weight files and checked the frozen list order and the checkpoint bindings. It did not regenerate the M8 estimator or the bootstrap and did not check the authenticity of receipts written on other machines. A second, independent pass regenerated the DiD and M8 point estimates, the 95 percent bootstrap endpoints, the bootstrap standard errors and the MDEs from the replacement rows, and each equals its saved value exactly as a binary64 number. Four qualifications remain. (i) The full-precision DoveNet M8 difference between the two scoring passes is not zero. The two-decimal displays agree. (ii) An alternate floating-point DiD computation agrees with the saved value within 10^{-9} and is not literally equal. (iii) The MDE file names the superseded evidence directory and records no digest of the script that wrote it, so the identity of that script is not established. Both DiD estimates and intervals in it match the replacement record, and the current script reproduces the file. (iv) The evaluation row files carry no individual byte hashes. These passes concern regeneration from the rows. They do not resolve the execution departures of main Section 8.

ccHarmony comparison. The values of main Section 4.5 and Table S8 come from the retained report, its manifest and an integrity-validation record. The report binds the manifest. The manifest does not bind the report bytes in return, which the frozen protocol did not require. One sentence describing P6 inside the saved report is out of date, and the agreement it reports uses both the retained and the official inputs. No source record or experiment was changed for this version of the paper.

S10 The identity-disjoint correction

Quantity	Value
Target photographs in both lists (train ∩ test)	1,399
of which HCOCO	1,391
of which HFlickr	8
Official test composites	7,404
Excluded test composites	1,823
path-identity exclusions	1,822
decoded-pixel duplicate	1
Retained test composites	5,581
Share of official test excluded	24.6 %
Official train composites (untouched)	65,742
Staged counts, train / val / test	52,411 / 13,331 / 5,581

Table S12: The identity-disjoint correction, derived by deterministic file-list arithmetic over the official release. Only the evaluation list changes; the train list is untouched, so published checkpoints remain eligible for rescoring without retraining.

Method	Official	Retained	Change
MSE ↓			
DoveNet	122.75	140.18	+17.43
Harmonizer	107.08	133.42	+26.34
gap (DoveNet − Harm.)	+15.67	+6.76	2.3× smaller
fMSE ↓			
DoveNet	1132.68	1114.71	−17.97
Harmonizer	794.24	894.34	+100.10
PSNR ↑ (dB)			
DoveNet	32.74	32.37	−0.37
Harmonizer	34.49	33.43	−1.06

Table S13: What changes if you adopt the list. The same checkpoints on the official and the retained list: MSE and PSNR are worse for both methods by unequal amounts, the gap between them shrinks by 2.3 times, and on fMSE the two move in opposite directions.

Worked example in the released lists. One target photograph, identifier `c344138`, carries three composites in the released lists: `c344138_1885590_1` at line 18,902 of the official training list, and `c344138_688383_1` and `c344138_309886_1` at lines 106 and 3,024 of the official test list. Both test composites are removed by the audit, the training composite stays, and the figure in the main text shows the first and the third of them sitting on the same photograph with a different recolouring.

List files and hashes. Three files are read, as the dataset repository distributes them: the official training list (65,742 rows, SHA-256 `d423036e...`), the official test list (7,404 rows, SHA-256 `513ebcf6...`) and the retained test list this release publishes (5,581 rows, SHA-256 `d2605e68...`). The full digests are in the project records.

S11 Soundness argument for the typed certificate

Setting and the two properties. Masked local editors (Blended Diffusion, DiffEdit, ZONE and BrushNet, cited in the main text) control which pixels are written, and compositing makes that exact. They do not state which outside pixels were allowed to steer the pixels written inside the region. We separate two properties. *Footprint locality* is the write-set property: the composite returns the source exactly where the effective write support is zero. It is not the same as the scoring region. Let $R_{\text{target}} = \{m > 0.5\}$ be the binary scoring region and let β be the feathered blend weight the composite applies. The composite zeroes the blend where β falls below a fixed feather threshold θ, so the effective write support is $R_{\text{write}} = \{\beta \geq \theta\}$, which can extend outside R_{target} because the mask is feathered, so a pixel outside R_{target} with $\beta \geq \theta$ can receive part of the edit. Compositing returns the source exactly outside R_{write}, and only there. *Sensitivity locality* is the causal property, and it is stated for the target coordinates inside the fixed trusted window.

Definition 1 (Sensitivity locality)**.** *E is (C, Ω)-sensitivity-local at (x, m, t) if, for the fixed scoring region $R_{\text{target}} \subseteq W$ and every perturbation δ supported on the exterior $\Omega = \neg W$ of the trusted window, $\|\pi_{R_{\text{target}}} E(x + \delta, m, t) - \pi_{R_{\text{target}}} E(x, m, t)\| \leq C\|\delta\|$.*

Theorem 1 (Typed certificate)**.** *Fix a mask m and trusted window W (the same for all inputs compared), let $R_{\text{target}} \subseteq W$ be the binary scoring region, and let the in-region output be*

$$\pi_{R_{\text{target}}} y = f_{\text{local}}(x^W, m) + |\gamma_P| \tanh f_P(x^W, m, g_P(x)) + |\gamma_S| \tanh f_S(x^W, m, g_S(x)),$$

where x^W agrees with x on W and is filled from x_W alone, the physical sub-channel is fed by the declassification map $P(x) = [\mu_{\text{RGB}}, \sigma_{\text{RGB}}]$ and the semantic sub-channel by a pooled feature, and the gates γ_P, γ_S are nonnegative (so $|\gamma_P| = \gamma_P$ and $|\gamma_S| = \gamma_S$). Assume finite weights, an evaluation-mode network of 3×3 circular convolutions with norm-preserving MaxMin activations on the stated square grid, and the statistics/nonlinear interface of the reported checkpoints. Then $L^P_{\text{cross}} \leq \gamma_P \sqrt{2}\, L(\text{enc}_P) L(\text{syn}_P)$ and $L^S_{\text{cross}} \leq \gamma_S\, L(\text{enc}_S) L(\text{syn}_S)$. In particular $\gamma_S = 0$ implies $L^S_{\text{cross}} \equiv 0$.

Soundness outline. The local path's out-of-window fill depends only on W, so its gradient with respect to x_Ω is zero. The mean pool's $1/\sqrt{N}$ and the broadcast's $\sqrt{N}$ cancel exactly, and per-layer operator norms use the FFT characterization of circular-convolution singular values (Sedghi et al., ICLR 2019), exact at the grid they are computed on, with norm-preserving MaxMin activations (Anil et al., ICML 2019). The $\sqrt{2}$ is the declassification factor and does not depend on the pooling size. The semantic certificate is zero by construction, a property of the computation graph rather than a selected outcome. Soundness is verified three ways in the repository: closed-form re-derivation, regression tests at odd resolutions with random Ω-supported perturbations, and a negative control that can fail. The full derivation, with norms, units and hypotheses, is Section S15.

Because the certificate upper-bounds how far admitted context can move the in-region edit, the bound can be compared with a declared context budget: opening the gate wider to force a larger in-region change raises L_{cross} and weakens the guarantee everywhere else. A budget-enforcing edit-or-refuse decision rule that acts on this comparison is not evaluated here. The only helper in the code chooses a gate by a quality threshold and, if none passes, returns the richest-context gate, which is not a fail-closed budget rule.

S12 Retrain design budgets

The registered final-checkpoint budgets are 25 epochs and 1,580,650 steps for DoveNet, and 60 epochs and 474,180 steps for Harmonizer. Each arm has 63,226 training composites: a common 61,264-composite unexposed base plus a matched 1,962-composite arm set. On the test side, exposed comprises 1,874 composites. Rest comprises 5,530 retained composites after removing 51 exposed overlaps.

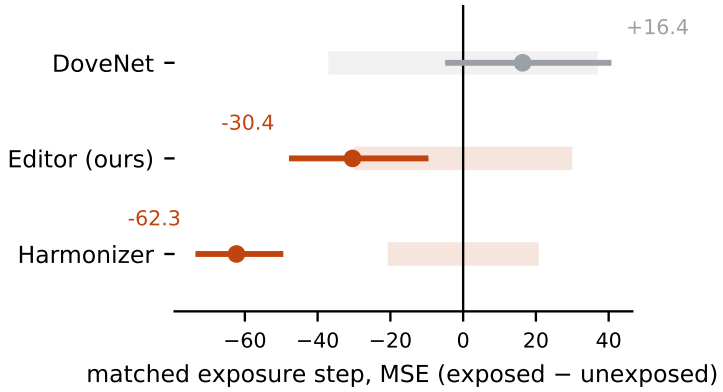


Figure S1: The matched exposure step with its detection floor. Points are the bin-averaged step, bars the 95 percent interval and the pale band the detection floor. Pooled rows only: Harmonizer's and the editor's intervals lie below zero; DoveNet's includes zero.

S13 Retrain departures per arm

This section records the departure classes and their claim consequences for the twelve retrain arms (Tables S14 to S17). In the repository the full-list arm is named `leak_in` and the reduced-list arm `leak_out`; the tables below use the repository names. Per-arm run roots, manifest fields, weight hashes, environment records and commit identifiers are kept in the project repository and are not reproduced here.

Class	Name	Meaning
X	Registered-item departure	Effective weight updates differ from the registered budget, whatever the manifest counter says
S	Stream continuity	Random stream or data order after a resume is not a continuation of the original, so the arm pair differs in more than exposure
E	Execution structure	Crash, pause, supervisor kill, separate invocations, uncollected attempts, co-tenancy
R	Runtime and environment	Machine, torch or python version, worker count, or recorded code revision differs between pairs
G	Provenance gap	A departure or tool is documented only in local-only files, or a hash or check output is missing
P	Post-tag collection	M8 scoring did not exist at the tag

Table S14: Departure classes used in Tables S15 to S17.

The values behind the departure classes come from the manifests and metrics sidecars of the collected arms.

Constant across the twelve arms, read from every manifest: the evaluation and training list hashes equal the registered ones, the confirmatory epoch equals the final epoch, and the scored

row counts are 5,581 and 1,874. The `is_pre_registered` flag is a driver-side flag; it is not evidence that the departures above did not occur.

Class	What the saved numbers may and may not be used for
X	The affected arms' fMSE rows may be reported as measured values at the stated checkpoint and lists, and may not be described as the product of the registered step budget or of a confirmatory arm.
S	A `leak_in` minus `leak_out` contrast involving a resumed arm may be reported as a difference between two training histories that includes exposure, and may not be described as an exposure-only difference or as a clean paired-seed contrast.
E	The manifests may be cited for what each arm recorded, and may not be cited as showing one uninterrupted invocation per pair or a driver-verified equal-steps check.
R	Pair-level numbers may be compared within a pair, and may not be pooled or compared across seeds as if run in one environment.
G	Facts that rest on local-only files may be stated as locally recorded, and may not be offered as shipped or independently reproducible evidence.
P	M8 may be used to say the trained checkpoints differ on composites that sit in one arm's training set and not the other's, and may not be described as a pre-registered check passed at the tag.

Table S15: Claim consequences per departure class, one sentence each.

Arm	Consequence
dovenet leak_in s0	The saved rows are a measured value with no recorded interruption; paired with leak_out s0 they support no exposure-only reading.
dovenet leak_out s0	The saved rows come from a reseeded run that applied about 2.1 percent more updates than registered; they are descriptive and are not a confirmatory arm.
dovenet leak_in s1	The saved rows come from a crash-and-resume run that applied about 1.0 percent more updates than registered and whose manifest was edited by hand to resume; they are descriptive only.
dovenet leak_out s1	The saved rows have no resume record but trained concurrently with its partner arm and a duplicate run; they support no claim of an isolated or uninterrupted invocation.
dovenet leak_in s2	Same as leak_in s0.
dovenet leak_out s2	Same as leak_out s0.
harmonizer leak_in s0	The saved rows are a measured value with no recorded interruption, trained in a different invocation from its partner.
harmonizer leak_out s0	The saved rows come from a reseeded resume from checkpoint 20 after a discarded first pass of about six epochs, preceded by an abandoned attempt; they are descriptive only.
harmonizer leak_in s1	The saved rows are an uninterrupted single-process run on laptop A, which does not carry over to its partner.
harmonizer leak_out s1	The saved rows come from three resumes, two without RNG restoration, with ten epochs of first-pass work discarded; they support no uninterrupted-RNG or exposure-only claim.
harmonizer leak_in s2	The saved rows are an uninterrupted single-process run on laptop B, which does not carry over to its partner.
harmonizer leak_out s2	The saved rows come from a resume at checkpoint 10 with recorded RNG restoration; bitwise continuity and an exposure-only contrast are not established.

Table S16: Claim consequences per arm, one sentence each.

Method	Consequence
DoveNet	All three pairs hold a class X arm, so the DoveNet DiD and its MDE are reported with neither difference nor equivalence established, and the pre-registered prediction "equivalent" is not supported.
Harmonizer	All three pairs hold a resumed or reseeded `leak_out` arm, so the Harmonizer DiD and its MDE are reported with neither difference nor equivalence established, and the pre-registered prediction "different, negative" is not supported.
Both	No method verdict, no causal null, no equivalence, no confirmed prediction; the word "confirmatory" is not used as a claim about either method.

Table S17: Claim consequences at the method level.

S14 Post-hoc sub-dataset check of the exposure step

Numbers behind the random-removal reference. Random removal of 1,807 HCOCO composites plus the 16 other excluded ones gives a remaining-set MSE gap of 2.07/3.69/5.14 and an official-to-remaining ratio of 3.05/4.25/7.57 at the 2.5th, 50th and 97.5th percentile, against the observed 2.32.

Label: post-hoc re-analysis of committed per-composite records, not pre-registered, CPU only, no model run. The retained record and tables use the estimator behind main Table 2, reused unchanged: eight quantile foreground-ratio bins computed on the records passed in, $k \geq 1$ minus $k = 0$ inside each bin, equal-weight mean over bins holding both arms, identity-cluster bootstrap, MDE = 2.8× bootstrap SE. Here the new strata use 2,000 draws. Main Table 2 pooled rows use 400 draws and were reproduced exactly at 400 draws (rng seed 11). A stratum is called estimable only with at least 30 unexposed and 30 exposed composites. PCT-Net, DucoNet and DiffHarmony are the registered-audit (E1, main Section 4.4) per-composite rows. DiffHarmony is outside the frozen four-method family and descriptive. No interval below is adjusted for multiplicity.

Method	Metric	Pooled step [95% CI]	HCOCO-only step [95% CI]	HCOCO MDE	Pooled minus HCOCO [95% CI]	Share
DoveNet	MSE	+16.36 [−6.4, +40.1]	+2.08 [−8.5, +13.4]	15.7	+14.27 [−2.4, +32.7]	87%
Harmonizer	MSE	−62.31 [−74.9, −49.7]	−2.86 [−7.0, +1.3]	5.9	−59.45 [−71.6, −47.6]	95%
Editor (ours)	MSE	−30.40 [−47.3, −10.7]	−2.32 [−10.0, +5.8]	11.0	−28.08 [−42.2, −11.4]	92%
PCT-Net	MSE	−42.49 [−52.0, −32.5]	−1.40 [−4.8, +2.1]	4.9	−41.09 [−49.5, −32.4]	97%
DucoNet	MSE	−69.35 [−82.4, −56.6]	−1.72 [−5.4, +2.1]	5.4	−67.63 [−79.8, −56.0]	98%
DiffHarmony	MSE	+50.23 [+41.0, +59.3]	+26.76 [+19.5, +34.3]	10.7	+23.47 [+16.5, +30.4]	47%
DoveNet	fMSE	+134.30 [+47.4, +223.1]	−40.50 [−120.2, +40.2]	115.0	+174.80 [+107.2, +240.2]	130%
Harmonizer	fMSE	−368.30 [−420.3, −314.2]	−50.33 [−82.5, −16.2]	46.2	−317.98 [−368.1, −270.8]	86%
Editor (ours)	fMSE	−188.46 [−251.5, −116.5]	−43.78 [−98.4, +10.4]	78.4	−144.67 [−200.4, −90.2]	77%
PCT-Net	fMSE	−272.49 [−313.2, −227.1]	−44.53 [−73.9, −15.9]	41.4	−227.96 [−264.0, −190.4]	84%
DucoNet	fMSE	−427.43 [−485.4, −369.5]	−35.65 [−65.7, −3.8]	44.0	−391.78 [−445.8, −340.1]	92%
DiffHarmony	fMSE	−44.58 [−84.0, −7.5]	−78.15 [−124.2, −33.4]	63.3	+33.57 [+2.4, +63.4]	−75%

Table S18: Pooled and HCOCO-only matched steps. Share is the pooled-minus-HCOCO difference over the pooled step. Inside HCOCO the steps of the three methods are not detected under this estimator and uncertainty calculation, so they are not read as an association with exposure.

Method (MSE)	Pooled retained / excluded [95% CI]	HCOCO retained / excluded [95% CI]
DoveNet	2.02 [1.75, 2.34]	1.47 [1.29, 1.69]
Harmonizer	5.05 [4.27, 5.88]	1.61 [1.41, 1.84]
Editor (ours)	3.06 [2.63, 3.55]	1.58 [1.38, 1.80]
PCT-Net	4.68 [3.95, 5.52]	1.56 [1.36, 1.78]
DucoNet	5.73 [4.82, 6.75]	1.56 [1.37, 1.77]
DiffHarmony	0.83 [0.79, 0.87]	0.90 [0.86, 0.93]

Table S19: Whole-image MSE ratios of the retained to the excluded list, pooled and inside HCOCO (identity-cluster bootstrap, 2,000 draws). The HCOCO ratios are not foreground-matched. DiffHarmony scores worse on the excluded list in both populations.

	Pooled MSE	HCOCO MSE	Pooled fMSE	HCOCO fMSE
Retained gap	6.76 [−5.20, +18.74]	58.73 [+52.38, +65.23]	220.38 [+166.64, +276.28]	710.70 [+667.26, +755.52]
Excluded gap	42.97 [+37.43, +49.26]	42.25 [+37.20, +48.11]	699.88 [+648.42, +752.37]	700.02 [+648.47, +753.15]
Official gap	15.67 [+6.44, +24.82]	51.77 [+47.55, +56.05]	338.44 [+294.60, +384.20]	706.19 [+669.98, +738.56]
Official / retained	2.32 [−13.42, +15.16]	0.88 [+0.84, +0.94]	1.54 [+1.37, +1.79]	0.99 [+0.96, +1.03]
Excluded − retained	+36.21 [+22.87, +50.25]	−16.48 [−24.56, −8.32]	+479.50 [+403.61, +557.22]	−10.68 [−77.97, +56.26]

Table S20: DoveNet-minus-Harmonizer gaps on the same composites. The pooled fMSE interaction is 479.50. Random removal of 1,807 HCOCO composites plus the other 16 gives an official-to-remaining ratio of 3.05 to 7.57 against the observed 2.32, a reference rather than a matched placebo.

Sub-dataset	Composites	Retained	Excluded	$k = 0$	$k \geq 1$
HCOCO	4,283	2,476	1,807	2,477	1,806
HAdobe5k	2,160	2,160	0	2,160	0
HFlickr	828	812	16	812	16
Hday2night	133	133	0	133	0
Pooled foreground bins (edges from the 7,404 composites): HCOCO share of the unexposed / exposed composites					
Bin 0 to 3	0.521/0.994, 0.756/1.000, 0.607/1.000, 0.472/0.985				
Bin 4 to 7	0.411/1.000, 0.420/0.989, 0.312/0.964, 0.220/0.925				

Table S21: Exposure by sub-dataset, one copy of each composite. HAdobe5k and Hday2night contain no exposed composite and HFlickr's 16 are below the 30-composite rule. Below, the largest-foreground bins carry most of the pooled step.

Scope. Nothing here is a causal statement. Exposed and unexposed composites inside HCOCO are not randomized. Among sub-datasets, only HCOCO passes the 30-composite reliability rule. The bins are the estimator's eight quantile bins with no other matching, and the 72 stratum records, of which 48 (all populations with at least one exposed composite) enter the H5 family and 24 (pooled and HCOCO only) pass the 30-unexposed/30-exposed rule above, carry no family-wise control at the per-bin level. The tables of this section fit no dose-response beyond $k = 1$ and pool the five seeds of the typed editor. Both were computed post-hoc by the statistics script of Section S16. No slope of error on k among exposed composites has an interval excluding zero for any of the three main Table 2 methods, in MSE or fMSE, under the specifications tried (linear and $\log_2 k$, equal-weight and fixed-effects, pooled and HCOCO bins, and bins 0 to 6 only). The detection floors are 8.6 to 56.4 MSE per extra copy, so this is an absence of detection and not a flat dose-response. Per seed (2,000 draws each), the pooled MSE step of the typed editor is -27.32 to -33.35 (five of five negative, every interval below zero, seed mean -30.40) and its HCOCO-only MSE step is -1.47 to -2.89 (every interval includes zero). The seeds repeat the pooled picture and are not independent composition estimates.

S15 Proof of the typed cross-region certificate

This section proves the inequality of the theorem stated in Section S11 for the code path that produced the reported checkpoints and states the hypotheses it needs. Every step follows the implementation's editor, Lipschitz and evaluation components. No reported value is changed. The statements concern the real-valued network and the float compositing path. The integer images of the audit are treated in S15.4.

S15.1 Setting *Scale.* The network works on floats in $[0, 1]$. The editor divides the uint8 image by 255, so a budget of ε levels (0 to 255) is $\varepsilon_{01} = \varepsilon/255$ and a network-scale quantity q

corresponds to $255\,q$ levels. The image is square, $n \times n$ with $n = 256$ (the audit passes the image height as the FFT grid), and $N = n^2$ is the pixel count. This N is not the $N = 1024$ samples per split of S6.

Norms. $\|\cdot\|_2$ is the Euclidean norm over all $3N$ entries (channels and positions), $\|\cdot\|_\infty$ the largest absolute entry. Every constant L below is an operator norm from $\|\cdot\|_2$ to $\|\cdot\|_2$ and is dimensionless.

Sets. $R_{\text{target}} = \{\text{mask} > 0.5\}$ with $n_R \geq 1$ pixels is the binary scoring region. The composite feathers the mask with blend weight $\beta \in [0,1]$ and hard preserves (blend set to zero) where the feathered weight falls below `outside_threshold` θ, so the effective write support is $R_{\text{write}} = \{\beta \geq \theta\}$, which can extend outside R_{target}. $W = R_{\text{target}}$ dilated by the halo (`dilate_window`, halo 0 gives $W = R_{\text{target}}$, the setting of every reported checkpoint). The statement covers target coordinates in the fixed trusted window W, and the exact footprint equality below holds only outside R_{write}. $\Omega = \neg W$ with $n_\Omega \geq 1$ pixels. The mask m, the halo and the feather settings are the same for all inputs compared. The composite defaults to feather radius 3 and outside threshold 0.01 (`cpu_editor.py`), and the wrapper settings of the reported checkpoints were not independently replayed here.

Perturbation class. For an image x, the admitted images are $x' \in [0,1]^{3\times n\times n}$ with $x'_W = x_W$, so $d = x' - x$ is supported on Ω. This contains every additive perturbation followed by clipping to $[0,1]$ (clipping cannot enlarge $|d|$). Changes inside W are outside the claim by construction.

The map. Let x^W be x with every Ω pixel set to the per-channel mean of x over W (the default fill of `GLWEditor.edit`). It is a function of x_W alone. Write $P(x) = [\mu_c(x), \sigma_c(x)]_{c\in\text{RGB}} \in \mathbb{R}^6$ with $\sigma_c = \sqrt{\text{var}_c + 10^{-12}}$, and bc for broadcast of a vector over the $n \times n$ grid. The residual is, as in `TypedGLWNet.forward`,

$$F(x) = f_{\text{loc}}([x^W, m]) + \gamma_P \tanh S_P([x^W, m, g_P(x)]) + \gamma_S \tanh S_S([x^W, m, g_S(x)]),$$
$$g_P(x) = \text{bc}\,\tanh E_P(P(x)), \qquad g_S(x) = \text{bc}\,\tanh\big(\tfrac{1}{N}\textstyle\sum_{\text{pos}} E_S(x)\big).$$

E_P is a stack of 1×1 convolutions with MaxMin on the $6 \times 1 \times 1$ map. f_{loc}, S_P, S_S, E_S are stacks of 3×3 circular convolutions with MaxMin. The edit is $\text{clamp}(x + F(x), 0, 1)$ followed by the feathered composite.

Hypotheses. (H1) The configuration is `physical_encoder = statistics`, `physical_interface = nonlinear` (the architecture stored in the reported checkpoints). The other encoder and interface kinds in the code are not covered. (H2) The network is in evaluation mode. It contains only convolutions, MaxMin and tanh, no normalization or dropout, `model_certified_lipschitz` raises an error for any other parameterized module. (H3) All weights are finite. (H4) The FFT constants are computed at the grid n of the input. They are not claimed for another size or for non-square images.

S15.2 Statement and proof Claim 1 (the declassified statistic). $\|P(x') - P(x)\|_2 \leq \|d\|_2/\sqrt{N}$ for all x, x'. The code uses the weaker constant $\sqrt{2/N}$ (`declassified_statistics`), which is also valid.

Proof. Fix a channel. Let $a, a' \in \mathbb{R}^N$ be its values, $\Pi = \mathbf{1}\mathbf{1}^\top/N$, $C = I - \Pi$. Then $\mu = \mathbf{1}^\top a/N$ gives $|\mu' - \mu| = \|\Pi(a' - a)\|_2/\sqrt{N}$. Also $\sigma = \|(Ca/\sqrt{N},\, 10^{-6})\|_2$ is a norm of an affine image of a, so the reverse triangle inequality gives $|\sigma' - \sigma| \leq \|C(a' - a)\|_2/\sqrt{N}$. Summing squares over channels and using $\|\Pi v\|^2 + \|Cv\|^2 = \|v\|^2$ gives $\|P(x') - P(x)\|_2^2 \leq \|d\|_2^2/N$. □

Claim 2 (layer norms). (a) tanh and MaxMin are 1-Lipschitz for $\|\cdot\|_2$ (MaxMin sorts disjoint channel pairs, and sorting does not expand the Euclidean distance). (b) A stack of 1×1 convolutions on a 1×1 map has Lipschitz constant at most the product of the spectral norms of

its weight matrices (`model_lipschitz_bound`). Call this $\mathrm{Lip}(E_P)$. (c) A 3×3 convolution with circular padding on the $n \times n$ grid is a circular convolution. Its operator norm is the largest singular value over the n^2 per-frequency transfer matrices (Sedghi et al., `conv_operator_norm_fft`), exact at that grid. The product over the layers of a stack bounds the stack (sub-multiplicativity). Call these $\mathrm{Lip}(S_P)$, $\mathrm{Lip}(S_S)$, $\mathrm{Lip}(E_S)$. Biases cancel in differences.

Theorem (network level). Under H1 to H4, for every admitted x',

$$\|F(x') - F(x)\|_2 \;\le\; (L^P + L^S)\,\|d\|_2,$$
$$L^P = \sqrt{2}\,|\gamma_P|\,\mathrm{Lip}(E_P)\,\mathrm{Lip}(S_P), \quad L^S = |\gamma_S|\,\mathrm{Lip}(E_S)\,\mathrm{Lip}(S_S),$$

and, entrywise, $|F(x') - F(x)| \le 2(|\gamma_P| + |\gamma_S|)$. These are the quantities returned by `cross_region_lipschitz_physical`, `cross_region_lipschitz_semantic` and their sum `cross_region_lipschitz`, with `input_size` $= n$. The bound holds on all positions, hence on R.

Proof. (1) *Local term.* $x'^W = x^W$ because $x'_W = x_W$ and the fill is a function of x_W, m is fixed. So f_{loc} contributes 0. (2) *Physical term.* The mean over a 1×1 map in `_broadcast_context` is the identity. By Claim 2, $\|\tanh E_P(P(x')) - \tanh E_P(P(x))\|_2 \le \mathrm{Lip}(E_P)\|P(x') - P(x)\|_2$. Broadcast over N positions multiplies the Euclidean norm by $\sqrt{N}$, so with Claim 1 in the code's form, $\|g_P(x') - g_P(x)\|_2 \le \sqrt{N}\,\mathrm{Lip}(E_P)\sqrt{2/N}\,\|d\|_2 = \sqrt{2}\,\mathrm{Lip}(E_P)\|d\|_2$. The factor $\sqrt{N}$ from broadcasting and the factor $1/\sqrt{N}$ from P cancel at every resolution. The input of S_P changes only in the g_P block, and $\mathrm{Lip}(S_P)$ is an operator norm over the whole input, so it bounds the change from that block alone. tanh is 1-Lipschitz. Multiply by $|\gamma_P|$. (3) *Semantic term.* For each of the 16 context channels, the mean over positions changes by at most $\|\Delta E_S\|_2/\sqrt{N}$ (Cauchy-Schwarz), so the pooled vector changes by at most $\mathrm{Lip}(E_S)\|d\|_2/\sqrt{N}$. After tanh and broadcast ($\times\sqrt{N}$) the factor $\sqrt{N}$ cancels again, giving $\|g_S(x') - g_S(x)\|_2 \le \mathrm{Lip}(E_S)\|d\|_2$. Then S_S, tanh and $|\gamma_S|$ as in step 2. (4) Add the two terms (triangle inequality). The entrywise bound is the range of tanh: each gated term moves an entry by less than $2|\gamma|$. □

Corollary 1 (edit, in-region). Let $e(x) = \mathrm{clamp}(x + F(x), 0, 1)$ and let $\beta \in [0,1]$ be the blend weight of the feathered composite. For $p \in R_{\mathrm{target}}$ the composite is $\beta(p)\,e(p) + (1 - \beta(p))\,x(p)$, and $x'(p) = x(p)$ because $R_{\mathrm{target}} \subseteq W$. Clamp is 1-Lipschitz and $\beta \le 1$, so $\|(\mathrm{out}(x') - \mathrm{out}(x))_{R_{\mathrm{target}}}\|_2 \le (L^P + L^S)\|d\|_2$ before integer rounding. Outside $R_{\mathrm{write}} = \{\beta \ge \theta\}$ the composite returns x exactly, which is the footprint statement. A pixel with $\beta \ge \theta$ that lies outside R_{target} can still receive part of the edit, and under an admitted perturbation the value of an Ω pixel can change, so R_{write} is not the theorem's target set.

Corollary 2 (budget in mean absolute units). If $\|d\|_\infty \le \varepsilon_{01}$ then $\|d\|_2 \le \varepsilon_{01}\sqrt{3n_\Omega}$. The mean absolute in-region change per pixel and channel is at most the root mean square, so with the per-channel tanh range as a second bound,

$$\bar{\Delta}_R \;\le\; b^P + b^S, \qquad b^\bullet = \min\Big(2|\gamma_\bullet|,\; L^\bullet\,\varepsilon_{01}\sqrt{n_\Omega/n_R}\Big).$$

This is `certified_influence_typed`, $255\,(b^P + b^S)$ is the audit column `cert255` and the quantity `mean_bound` of the tightness check (times 255). The geometric factor $\sqrt{n_\Omega/n_R}$ converts the Euclidean budget of the theorem to the per-pixel mean the audit measures. It is large when the region is small, which is why the table below is far from $L\varepsilon_{01}$.

S15.3 Zero semantic term and physical dependence (i) With $\gamma_S = 0$ exactly, the semantic output term is multiplied by 0 (finite tanh values, H3), so $L^S = 0$ and no path leads from Ω through the semantic modules. This is a statement about the computation graph: the

unrestricted learned-context branch is disabled and does not influence the output, and nothing about that branch is certified. It does not imply that exterior semantic content is absent. The channel labels are implementation names. The remaining exterior dependence is six global color moments, and those moments can correlate with semantic content, so this is not a disentanglement claim. In the five reported checkpoints `L_cert_semantic` is 0.0. (ii) The physical channel reads Ω only through $P(x)$. If $P(x') = P(x)$ the physical term is identical: any permutation of Ω pixel values leaves every per-channel mean and standard deviation unchanged. The regression test `test_declassified_channel_is_invariant_to_spatial_semantic_rearrangement` checks the statistic to 10^{-7} and the edited region exactly. In float arithmetic equality holds only up to summation order. If $P(x') \neq P(x)$ the output can move, by at most $L^P \|d\|_2$: a uniform shift of Ω changes the channel means and therefore the edit. So at $\gamma_S = 0$ the editor is not independent of Ω. It depends on six numbers, with the stated bound. (iii) At $\gamma_P = \gamma_S = 0$ the output is independent of Ω (test `test_typed_local_write_ignores_omega_when_global_gates_zero`).

S15.4 Code checks and scope *Pool and broadcast.* The mean-pool/broadcast cancellation of step 3 applies to the semantic channel. The physical channel has a different cancellation: broadcast $\sqrt{N}$ against the statistic's $1/\sqrt{N}$, leaving $\sqrt{2}$ with the code's constant. *No downsample factor.* The current source contains no `avg_pool2d`. The pooled low-pass encoder whose factor $1/k$ cancelled $\sqrt{N/N_p} = k$ was retired, `phys_downsample` is kept in the constructor and checkpoint metadata only and is not read by the forward pass. The certificate therefore does not depend on a downsample factor. *Constant.* By Claim 1 the factor $\sqrt{2}$ is valid but not tight, 1 would also be valid. It is not applied here and no reported value uses it. *Slack in* $\mathrm{Lip}(S)$. It is the norm over the whole input, an upper bound for the g block. *Resolution.* The FFT norm is exact only at the grid it is computed on, so the constants are stated at $n = 256$ only, and it is not monotone in the grid size in general: non-nested frequency grids can miss different maxima, so a norm at one size is not a bound at another. A scalar kernel whose only nonzero row is $[1, -1, 0]$ has circular operator norm 2 at $n = 4$ but $2\cos(\pi/10) < 2$ at $n = 5$. Certification is therefore required at the resolution actually evaluated. *Integer rounding.* The editor truncates to uint8 twice (the edited image and the composite) and the audit truncates the perturbed input. Truncation can move a value by up to one level while the real-valued change is arbitrarily small, so audit columns measured on uint8 outputs can exceed $255\,(b^P + b^S)$ by about one level per truncation. The corresponding regression test allows a margin of one level. The projected-gradient tightness check measures the float network residual and has no such margin. *Arithmetic.* The proof is for real arithmetic. The constants are float32 FFT and singular-value computations with no outward rounding, so the computed value is an estimate and not an interval-certified machine bound, and we state no numeric rounding allowance here. A CPU recomputation from the saved checkpoints reproduced the saved `mean_bound` values to a relative difference below 10^{-5}. *Role of PGD.* The proof carries soundness. The search (S5) is a consistency check of the code against the proof, and with about 13× slack it cannot detect a bound that is wrong by a small factor. *Not covered.* Halo values other than 0 were not part of the reported runs. The proof does not use the halo value. Nothing is claimed for editors outside H1 or for perturbations that touch W.

S15.5 Units for the reported checkpoints Table S22 applies Corollary 2 to the five selected checkpoints. L^P is the selected point's `L_cert_physical` (the range of Table S9). $8L^P$ is $255\,L^P\varepsilon_{01}$ for $\varepsilon = 8$ levels, the value before the geometric factor. The next column is the saved `mean_bound` of the tightness check times 255, a mean over 100 inputs of the retained list. The last column is its ratio to $8L^P$. Where the tanh cap $2|\gamma_P| = 0.2$ (51 levels) binds for some input,

the mean falls below the product and the ratio falls below the common value of the two uncapped seeds (4.73, the mean of $\sqrt{n_\Omega / n_R}$ over those 100 inputs). The edit magnitude is the validation mean absolute in-region edit of the selected point. It comes from a different input set than the bound.

Seed	L^P	$8L^P$	Mean bound (levels)	Bound / $8L^P$	Edit magnitude (levels)
0	0.4170	3.34	15.37	4.61	9.43
1	0.0758	0.61	2.87	4.73	8.98
2	0.0489	0.39	1.85	4.73	8.06
3	0.4236	3.39	15.60	4.60	9.19
4	0.4866	3.89	17.70	4.55	9.60

Table S22: Certified in-region bound at eight levels, mean absolute units per pixel and channel, five selected checkpoints. The bound is below the edit magnitude for the two seeds with the smallest bound and above it for the other three.

S16 Post-hoc multiplicity sensitivity

The within-HCOCO fMSE residuals are post-hoc sensitivity results, not evidence for the main claim. H1 is the 12 within-HCOCO method-by-metric tests. H5 is the broader family of all 48 estimable stratum records. The frozen four-method family excludes DiffHarmony, which is shown only descriptively. The two families use the same raw bootstrap p values and different Holm adjustments. Harmonizer and PCT-Net survive H1, while neither survives H5. Table S23 uses 20,000 bootstrap draws and Table S18 uses 2,000, so the two tables give the same point estimates with interval endpoints about one unit apart (Harmonizer fMSE −50.33: $[-83.3, -17.4]$ here, $[-82.5, -16.2]$ in Table S18). No sign changes.

Method	Step	95% CI	Raw p	H1 Holm	H1	H5 Holm	H5
DoveNet	−40.50	$[-122.0, +41.2]$	0.3311	1.0000	no	1.0000	no
Harmonizer	−50.33	$[-83.3, -17.4]$	0.0028	0.0252	yes	0.1008	no
Editor (ours)	−43.78	$[-99.0, +11.3]$	0.1242	0.8694	no	1.0000	no
PCT-Net	−44.53	$[-73.6, -15.2]$	0.0024	0.0240	yes	0.0912	no
DucoNet	−35.65	$[-66.9, -4.7]$	0.0240	0.1920	no	0.7920	no
DiffHarmony (descriptive)	−78.15	$[-121.8, -35.8]$	0.0006	0.0066	yes	0.0234	yes

Table S23: Within-HCOCO fMSE exposure steps. H1 is the 12-row within-HCOCO family. H5 is the 48-row all-estimable-strata family. DiffHarmony is outside the frozen four-method family and is descriptive only. Values were regenerated by the committed statistics-hardening script.

S17 Harmonizer reconciliation

The main text reports Harmonizer under one protocol throughout, the protocol of main-text Section 4: every composite, mask and target resized to 256×256, arguments predicted on that resized composite. Its own paper reports the benchmark at the same resolution (arXiv 2207.01322, Table 1), but its evaluation code predicts the filter arguments from the native-resolution composite and resizes only afterwards. Running its protocol, with its released model code and its own metric function, on the staged composites gives the published row.

Quantity	Their protocol, our run	Published	Difference
MSE (official list)	23.99	24.26	1.10 percent
fMSE (official list)	277.02	280.51	1.24 percent
PSNR (official list)	37.88	37.84	0.04 dB
MSE, our protocol	107.08	n/a	their number under our setting

Table S24: The Harmonizer checkpoint is the one its paper describes. The 107.08 quoted in the main text's tables is that same checkpoint under our protocol, and the difference is the resolution at which the arguments are predicted, not a different model.

The same choice moves the population contrast for this checkpoint. Under our protocol the retained-to-excluded ratio is 5.05 on MSE and 1.83 on fMSE. Under its own it is 1.99 and 0.99, so the apparent inflation is smaller under its authors' setting and absent on the foreground-region metric. The population tables of the main text stay in one protocol, and this is the stated sensitivity to it. Runs: three populations, eight CPU threads, about three hours wall clock, dominated by the high-resolution sub-datasets because the argument prediction runs at native resolution. Records: the summary files of the reconciliation under the release's notes, each with per-subset and pooled values, per-sample rows and the published row it is compared against.

S18 Metric validation

The dataset README reports two DoveNet rows: the paper model at MSE 52.36, PSNR 34.75 dB and fMSE 549.96, and the released checkpoint we score at an expected MSE 50.87 and PSNR 34.83 dB, with no fMSE. Our pinned official-list numbers score the released checkpoint, so the 52.36 row is not the reference for them. Our PSNR averages per-image PSNR at a fixed peak of 255, as the published tables do, while the upstream scripts use each image's own output range as the peak, which reads lower on the same images: the same list under that convention gives MSE 124.50 and PSNR 32.04 dB.

Data path. Scoring the unharmonized composite with our loader and metrics over all 7,404 official composites gives MSE 170.06, fMSE 1341.41 and PSNR 31.96 dB against the README's 172.47, 1387.30 and 31.63 dB, with per-subset MSE within 2.6 percent and per-subset PSNR within 0.6 dB (largest deviations: HCOCO MSE −2.5 percent and Hday2night PSNR +0.56 dB). This fixes the test list, file pairing, resizing, colour handling, region of integration and pixel scale as matching the published protocol.

BatchNorm mode. The upstream test script, run as its README documents, never enables evaluation mode and forces batch size one, so its batch normalization uses per-image statistics. Our driver uses inference-mode statistics. Table S25 reruns the released checkpoint over the full official list under that upstream protocol (uint8 truncation, PSNR peak equal to each image's output range).

Subset	n	our MSE	README	ratio	our PSNR	README PSNR	diff
HCOCO	4,283	33.61	33.65	0.999	35.98	35.95	+0.03
HAdobe5k	2,160	56.40	55.99	1.007	34.23	34.26	−0.03
HFlickr	828	125.92	125.73	1.002	30.58	30.58	+0.00
Hday2night	133	56.97	56.26	1.013	34.73	34.76	−0.03
overall	7,404	51.00	50.87	1.003	34.84	34.83	+0.01

Table S25: Released-checkpoint reproduction under the upstream protocol. The README column is the released model's documented expectation, and the whole gap between our printed numbers and the published row is the batch-normalization mode.

Population contrast under both conventions. The contrast the paper reports is present in both modes and larger under the upstream one. Table S26 gives the pooled populations and the HCOCO restriction in inference mode (columns "eval") and training mode (columns "train").

Pooled population	n	MSE eval	MSE train	fMSE eval	fMSE train
official (all)	7,404	122.75	49.99	1132.68	537.53
retained	5,581	140.18	58.21	1114.71	538.88
excluded	1,823	69.40	24.83	1187.67	533.39
ratio retained / excluded		2.02	2.34	0.94	1.01

Inside HCOCO	n	MSE eval	fMSE eval	MSE train	fMSE train
retained	2,476	98.96	1240.67	38.75	554.04
excluded	1,807	67.16	1181.79	24.46	532.71
ratio retained / excluded		1.47	1.05	1.58	1.04

Table S26: Population contrast under inference-mode ("eval") and training-mode ("train") batch normalization, from the full-list run whose replication control passed (max $|\Delta\text{MSE}|$ 5.4×10^{-5} against the stored records). The official list is a weighted average over two differently composed populations: the excluded list is 99 percent HCOCO, the retained 44 percent. Nothing here licenses a causal exposure claim.

Only the population-level ratios of Table S26 are shown to hold under both conventions; the foreground-ratio-matched step contrasts of main Table 2 were not recomputed in training mode.